

FinGear AI: An Explainable Financial Digital Twin for Financial Health Scoring, Anomaly Detection and Forecasting

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Abstract— FinGear AI is a personal financial intelligence tool, built as a Financial Digital Twin. It is intended to support personalized financial health, optimize decision-making and improve financial discipline. Users can safely save financial profiles, log income and expenses, upload invoices in PDF format and monitor transactions. Rather than strictly adhering to a rigid 50:30:20 rule, the platform categorizes users by income brackets to apply flexible, income-based budgeting ratios that track spending across needs, wants, and savings. It evaluates spending behaviour, cash reserves, emergency funds, debt metrics, saving habits, investment consistency and payment histories to generate a transparent, interpretable score on financial wellness together with proper justifications, risk alerts and practical suggestions. Furthermore, anomalous transactions are detected, allowing users to confirm valid records or exclude abnormal ones from model training. Customized spending and cash-flow projections are built on rule-based estimates for new users that dynamically evolve to custom Random Forest and ARIMA models as transaction data accrues. FinGear AI is built on a stack of React, FastAPI, PostgreSQL and machine-learning services to provide a secure, transparent and scalable platform to make better personal financial decisions.

Keywords— Financial Digital Twin, explainable artificial intelligence, financial wellness score, financial discipline, anomaly detection, ARIMA, Random Forest regressor, personal finance management, income-based budgeting ratios.

I. INTRODUCTION

Financial wellness depends on a person's ability to meet current commitments, handle financial shocks, manage debt and progress toward future goals. These outcomes are not visible from income alone; a high-income individual may face unstable cash flow or inadequate reserves, while a lower-income individual may be resilient through disciplined behaviour. A useful personal-finance system must therefore analyse behaviour in context rather than impose a single generic rule.

Existing personal-finance applications largely record income, expenses and balances, offering fragmented and reactive insight. Users may see how much they spent in a category without understanding whether the pattern is sustainable, or how it affects their emergency fund, debt position or long-term goals. Generic rules such as the fixed 50:30:20 allocation cannot account for differences in income, dependents, debt obligations and financial priorities.

FinGear AI addresses this gap through an explainable Financial Digital Twin. This paper reports the Phase I assessment layer of the system, which is limited in scope to three tightly connected modules: (i) an adaptive, explainable Financial Health Score, (ii) category-specific transaction anomaly detection, and (iii) progressive expense and cash-flow forecasting. Each output is derived from data the user has supplied, is traceable to an explicit rule or model, and is designed to support the later Digital Twin, planning and advisory features of the system.

II. LITERATURE REVIEW

Uyanahewa et al. proposed WONGA, a smart personal-finance application that automates expense tracking through bill-image processing, bank-SMS extraction, natural-language processing and calendar-event expense prediction. The system used ANN models for SMS classification, OCR and NER for bill extraction, Multinomial Naive Bayes for expense categorisation, ARIMA for short-term expense forecasting, and DistilBERT with regression models for event-based expense estimation, reporting 97% SMS classification accuracy, 88% transaction-extraction accuracy and 98.93% expense-classification accuracy. Its limitations include dependence on region-specific datasets and the absence of an explainable health score, adaptive income-tier budgeting, anomaly feedback or Digital Twin modelling [1].

Hyndman and Khandakar presented automatic forecasting with exponential smoothing and ARIMA, providing the

statistical basis for FinGear AI's forecasting pipeline. Their work is general-purpose and does not address personal-finance context, transaction classification or explainable recommendations [2].

Anshari, Almunawar and Masri explored Financial Digital Twins for robo-advisory, showing how personal data and analytics can support dynamic, customised advice. The work is conceptual and does not implement a complete transaction-level health-scoring and forecasting system [3].

Bonelli studied Digital Twin personalisation and conversational AI in robo-advisory using a scenario-based randomised experiment with 336 respondents. Digital Twin personalisation improved perceived personalisation but increased privacy concern, while conversational AI improved perceived interaction quality; the study did not evaluate real financial outcomes [4].

Acharya and Bhojak proposed a Financial Doppelgangers framework combining Digital Twins, Graph Neural Networks, Distributed Ledger Technology and Explainable AI for systemic-risk detection in FinTech ecosystems. Evaluated on over 12,000 peer-to-peer lending transactions, it reported 88% early-warning accuracy and AUC 0.91, but targeted ecosystem-level rather than individual personal-finance management [5].

Jiang et al. proposed an LSTM and multi-objective optimisation approach for personalised investment advice using market information, behaviour, goals and risk appetite. Simulated results showed more stable responses to market volatility, but the study did not address real-world suitability or integration with everyday spending behaviour [6].

Kong et al. reviewed Large Language Model applications in finance, covering financial text analysis, sentiment analysis, forecasting, reasoning and agent-based modelling. They highlighted the potential for personalised advice alongside hallucination, reasoning and privacy risks, supporting the use of structured financial data to ground explanations [7].

Das, Markowitz, Scheid and Statman developed a mental-account portfolio framework linking Mean-Variance Theory, Behavioural Portfolio Theory and goal-based risk management. The work demonstrates mathematical treatment of goal-specific portfolios but does not incorporate real-time personal transactions or AI-based behaviour analysis [8].

Leukam, Koffi and Djagba proposed goal-based portfolio optimisation using G-Learning and inverse reinforcement learning, reporting an improvement in simulated Sharpe Ratio from 0.42 to 0.483. The evaluation is simulation-based and does not fully account for real transaction costs, market impact or consumer deployment [9].

Feng, Li and Liu proposed a responsible-AI framework for robo-advisory and financial planning centred on fiduciary

duty, adaptive personalisation, robustness, ethics, fairness and auditability. The work provides a conceptual roadmap rather than a fully implemented financial-planning system [10].

Černevičienė and Kabašinskas reviewed 138 studies on Explainable AI in finance, identifying credit management, stock prediction and fraud detection as common applications and feature importance, SHAP and rule-based methods as frequent explanation techniques. The review also identifies unresolved challenges in explanation quality and fairness [11].

Collectively, the literature provides separate foundations for transaction processing and ARIMA forecasting [1], automatic forecasting [2], Financial Digital Twins and trust [3], [4], [5], goal-oriented investment [6], [8], [9], and Explainable AI [11]. No reviewed system combines adaptive income-tier budgeting, a connected transaction ledger, an explainable health score, per-user category-specific anomaly feedback and progressive forecasting in one architecture; Phase I of FinGear AI addresses this gap.

III. METHODOLOGY

As shown in Fig. 1 below, the Phase I architecture comprises a React frontend along with a FastAPI backend and a PostgreSQL database, and is divided into two interconnected blocks, i.e., Financial Analytics System and Personalized Decision Support layer. Data related to financial profile and transactions, which is provided via the frontend interface, goes through Application Programming Interface (API) routes authenticated via Json Web Token (JWT). After this, the data processing pipeline validates the data, categorizes the transactions and prepares the features in accordance with user history. The PostgreSQL database acts as the single source of truth for profiles, transactions, history of health scores, anomaly feedback and predictions.

Three engines work with the common financial environment: an anomaly detection engine that detects new expenses in comparison with the user's existing category-based behaviour; a forecasting engine based on a statistical baseline and an Autoregressive Integrated Moving Average (ARIMA) model, as well as a personalized Random Forest regressor depending on the level of historical data; and the financial health engine that applies the rules for calculating financial health based on the classic 50:30:20 principle modified for different income levels, as seen in Table I. These engines send their results via API to the Personalized Decision Support module that constructs transaction and expense tracking interfaces and health and forecasting dashboards, along with the goal setting, what-if scenario simulation, AI-Copilot assistance and investment decision support interfaces in future phases. This architecture implies that the Financial Digital Twin is a logical construct over the shared analytical environment and not a separate database.

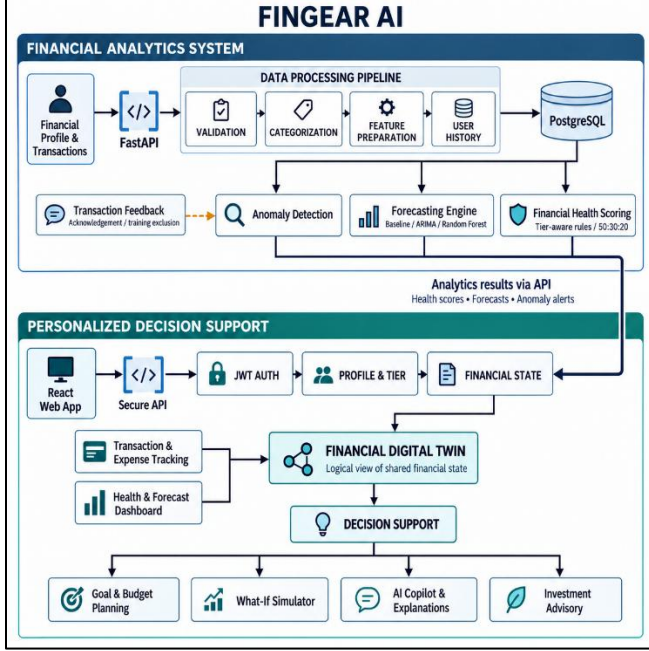


Fig. 1. FinGear AI system architecture (Financial Analytics System and Personalized Decision Support)

TABLE I. INCOME TIERS AND TARGET ALLOCATIONS

Income Tier	Monthly Income	N:W:S (%)
Survival	Below ₹30,000	65:15:20
Baseline	₹30,000–₹49,999	50:30:20
Accumulation	₹50,000–₹79,999	45:25:30
Reverse Budget	₹80,000–₹1,24,999	40:20:40
Wealth Building	₹1,25,000 and above	35:15:50

A. Financial Health Score

Needs–wants–savings budgeting [12], the EMI-to-income (debt-to-income) ratio [13] and emergency reserves sized against essential expenses [12], [14] are established personal-finance principles. The contribution of FinGear AI is not new financial mathematics but their adaptive, explainable integration: targets vary with income tier, and the indicators are combined with safety guardrails into one transaction-driven score.

Users are classified into the five income tiers of Table I, each with a distinct Needs–Wants–Savings (N:W:S) target ranging from 65:15:20 for Survival to 35:15:50 for Wealth Building.

Let I denote average monthly income, NE needs expenditure, WA wants expenditure, SV savings, EMI mandatory debt repayment, OD outstanding debt and EF liquid emergency funds; N_T , W_T and S_T are the tier's target ratios. SS , ER , DM , FR and PD denote Spending Sustainability, Emergency Resilience, Debt Manageability, Future Readiness and Payment Discipline and Protection. The actual allocation ratios are then computed as in (1):

$$N = NE / I, \quad W = WA / I, \quad S = SV / I \quad (1)$$

Spending Sustainability (weight 30%) combines allocation fit and surplus fit. NeedsFit, WantsFit and SavingsFit measure the normalised deviation of the actual allocation ratios from the tier targets, scoring 100 when needs and wants are at or below their targets and savings at or above, as in (2); SurplusFit measures the achieved monthly surplus $MS = I - NE - WA$ relative to the target savings amount $TS = S_T \cdot I$, as in (3). They are combined using (4) and (5):

$$\begin{aligned} NeedsFit &= \min(100, 100 \cdot N_T / N) \\ WantsFit &= \min(100, 100 \cdot W_T / W) \\ SavingsFit &= \min(100, 100 \cdot S / S_T) \end{aligned} \quad (2)$$

$$SurplusFit = \min(100, 100 \cdot \max(MS, 0) / TS) \quad (3)$$

$$AllocationFit = 0.25 \cdot NeedsFit + 0.30 \cdot WantsFit + 0.45 \cdot SavingsFit \quad (4)$$

$$SS = 0.75 \cdot AllocationFit + 0.25 \cdot SurplusFit \quad (5)$$

Emergency Resilience (weight 25%) compares liquid funds against a risk-adjusted target, as defined in (6). EmergencyTarget is the essential monthly expenditure NE multiplied by a reserve period of 3 months for salaried users without dependents, 4 months with dependents, 5 months with more than two dependents and 5 to 6 months for self-employed, freelance or business users, with a minimum floor of ₹25,000 for Survival-tier users:

$$ER = \min(100, 100 \cdot EF / \max(EmergencyTarget, 1)) \quad (6)$$

Debt Manageability (weight 20%) penalises high monthly debt-service pressure and total leverage. With $r_E = EMI / I$ and $r_D = OD / (12 \cdot I)$, the sub-scores are given in (7) and the component in (8):

$$\begin{aligned} EMIScore &= \min(100, \max(0, 100 \cdot (0.40 - r_E) / 0.25)) \\ DebtScore &= \min(100, \max(0, 100 \cdot (1.50 - r_D))) \end{aligned} \quad (7)$$

$$DM = 0.60 \cdot EMIScore + 0.30 \cdot DebtScore + 0.10 \cdot (OnTimeRate \cdot 100) \quad (8)$$

EMIScore is 100 when EMI is at most 15% of income and falls linearly to zero at 40%; DebtScore falls linearly from 100 to zero as outstanding debt rises to 1.5 times annual income; OnTimeRate denotes the percentage of on-time repayment to total number of repayment of debts.

Future Readiness (weight 15%) evaluates saving-target achievement, goal feasibility and investment consistency, as given in (9):

$$FR = 0.50 \cdot SavingTargetAchievement + 0.30 \cdot GoalFeasibility + 0.20 \cdot InvestmentConsistency \quad (9)$$

SavingTargetAchievement is the achieved savings SV as a percentage of the target TS , capped at 100. GoalFeasibility averages, over active goals, the planned monthly contribution as a percentage of the required one (100 when met, otherwise 10–100). InvestmentConsistency is the share of the last three months with an investment contribution. With no active goal, the goal weight is redistributed proportionally.

Payment Discipline and Protection (weight 10%) evaluates on-time payments, planning status and insurance coverage, as given in (10):

$$PD = 0.60 \cdot \text{PaymentScore} + 0.25 \cdot \text{PlanningScore} + 0.15 \cdot \text{ProtectionScore} \quad (10)$$

PaymentScore is the on-time payment rate scaled to 100. PlanningScore is 100 when a budget, emergency planning and active goals are all in place, 65 when any two are, and 30 otherwise. ProtectionScore is 100 when insurance expenses are recorded and 50 with only provident-fund or fixed-deposit information.

The overall raw score combines all five weighted components as shown in (11), and the guardrails in (12) produce the final Financial Health Score (FHS), which is rounded and restricted to the range 0–100:

$$\text{RawScore} = 0.30 \cdot SS + 0.25 \cdot ER + 0.20 \cdot DM + 0.15 \cdot FR + 0.10 \cdot PD \quad (11)$$

$$FHS = \min(\text{RawScore}, C_s, C_e) \quad (12)$$

Here $C_s = 59$ when $MS < 0$ and 100 otherwise, and $C_e = 49$ when EMI reaches 40% of income and 100 otherwise. Both these safety limits ensure that the good performance in one category cannot mask a bad affordability or debt situation. The weights are design parameters and not statistical figures. This means that the engine gives priority to sustainable expenditure and ability to meet emergencies (55% collectively). It takes into account the scores on debt, planning for future, and payment discipline without evaluating their sensitivity. Score ranges from Financially Healthy (80-100), Stable/ Improving (60-79), Needs Attention (40-59), and Financially Vulnerable (0-39). Apart from scoring the financial situation of the user, the engine also triggers risk alerts whenever there is less than one-month reserve for emergencies, any unpaid bill or EMI in the past 90 days, EMI pressure exceeding 35% of the income, or debt exceeding 1.2 times annual income.

There is another data confidence level, defined in (13). This is independent from the health score calculation and thereby takes account of incomplete information without implicitly decreasing the user's score:

$$\text{DataConfidence} = 0.40 \cdot \text{HistoryCompleteness} + 0.40 \cdot \text{CategorisationCompleteness} + 0.20 \cdot \text{IncomeVerification} \quad (13)$$

The system indicates High Confidence when the data confidence level exceeds 80%, Moderate Confidence for scores between 50% and 79%, and Provisional Score for data confidence levels lower than 50%. Finally, the recommendations system turns the score into actions: excessive discretionary spending is trimmed gradually, not in one shot. Hence, the proposed decrease in expenses does not exceed a third of the excessive spending or 5% of income, whichever is less.

B. Transaction Anomaly Detection

Each user has an isolated anomaly detection model for every spending category like Food, Transport, Housing, Utilities, Shopping, Entertainment, and many others. Hence, same transaction amount could be normal in one category and abnormal in another category for the same user. If there is not enough history for a certain category for a particular user, a special cold start procedure is applied. With this, the anomaly could be evaluated even in the absence of historical data. With further accumulation of transaction data, the model switches to a machine learning algorithm. This includes a combination of K-Means clustering for the category and autoencoders for anomaly detection.

The module provides information about the spending category for each transaction, its normalized anomaly score and severity level. It also displays the maturity status of the current model and a brief description of the anomaly's cause. Unlike other modules described above, this module does not automatically block the transaction. Instead, it acts as an additional layer to the decision-making process. The user may confirm the transaction is normal and this transaction will become a part of the baseline. Alternatively, he/she could exclude the transaction from model's training in the future, and this transaction will remain visible in the ledger, but will not affect the learning process anymore. In this way, a unique event such as an invoice for an urgent doctor visit will not change the model's perception of the user's behaviour in the future. Furthermore, because the models are isolated for each user and each spending category, one user's behaviour will not affect the baseline of another user's models.

C. Progressive Expense Forecasting

The forecast engine selects the model dynamically depending on the quantity and quality of historical data for each individual. This happens through three successive phases instead of just picking a single model for everyone. The first model to be applied is the statistical/rule-based model. This is used in cases where there is no historical data for the user, fewer than seven distinct expense days in the category, or when historical data is insufficient. It gives a quick estimation based on past averages and frequencies. Moreover, it includes a maturity value so the user knows the level of confidence associated with it. If there are more than seven distinct expense days, a personalized Random Forest regression model is created by training it on the transactions of the user. It detects any non-linearities in their spending behaviours that cannot be caught using an average.

With the gathering of multi-month time series data of a user, an ARIMA model is further trained. It accounts for trend and temporal dependency between consecutive periods. This helps in supplementing the capabilities of the Random Forest algorithm, which models the relationship among multiple attributes non-linearly. Before modelling is done, the raw transaction data is pre-processed into features such as

date, type, category, amount, day/week/month, recurrence and recent trend features. It is based on the validated ledger of the health-score and anomaly detection modules. Daily predictions help in making short-term plans as they give the predicted expense of the coming days. Monthly and category-wise predictions help in making projections on cash flow movement, as described in (14) and (15):

$$ProjSurplus = ProjIncome - ProjNeeds - ProjWants - ProjDebtPayments \quad (14)$$

$$ProjSavingsCapacity = \max(0, ProjSurplus) \quad (15)$$

Every prediction is designated as provisional, forming or mature based on the extent of transaction history for the user, number of different days of expenses, category coverage, and stability of the spending pattern. This is done so that any prediction generated based on few days' worth of data can never be given as much certainty as that of a prediction generated from an established monthly data series. Predictions are automatically recomputed anytime a new transaction, modification of a recurring transaction, income changes, recovery of a deleted entry, or a change in the training data set occurs.

IV. RESULT AND ANALYSIS

The three Phase I modules were evaluated independently using module-specific benchmarks.

A. Financial Health Score

This evaluation was conducted across all the five categories of income groups and the implementation scored 100 percent compliance to the pre-specified scoring criteria for the scenarios examined. Table II gives some representative results. The other criterion which scored 100 percent is the critical debt guardrail criterion where, when the ratio of EMI to income is equal to or greater than 40 percent, the score should be capped at 49 or lower.

TABLE II. HEALTH SCORE VALIDATION ACROSS INCOME TIERS

Income Tier	Monthly Income	Score	Assessment
Survival	₹22,000	38/100	Vulnerable
Baseline	₹42,000	72/100	Improving
Accumulation	₹72,000	92/100	Healthy
Reverse Budget	₹1,15,000	97/100	Healthy
Wealth Building	₹2,40,000	97/100	Healthy

B. Transaction Anomaly Detection

This ensemble was tested on 300 transactions that had been labelled (240 transactions with normal behaviour and 60 with anomalous behaviour) and obtained 95.67% accuracy rate. The total number of normal transactions (240 transactions) was correctly classified, while 47 out of 60 anomalous transactions were found, which implies that the precision value is 100%, the recall is 78.33%, the F1-score is 0.8785, and the Receiver Operating Characteristic – Area Under the Curve (ROC-AUC) is 0.9992. There were no false positives, which means that all flagged transactions had the correct

label (they were indeed anomalous), but there were still 13 anomalous transactions unclassified as anomalous, as seen from Fig. 2 and Fig. 3.

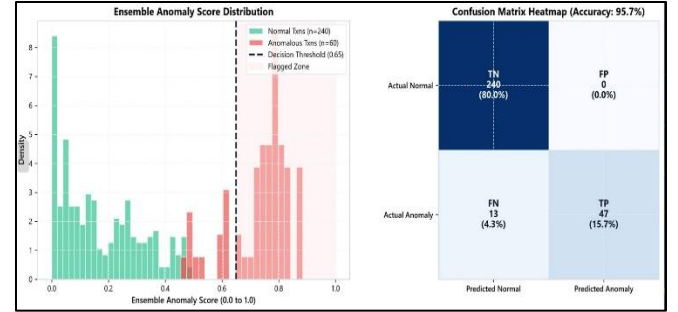


Fig. 2. Anomaly score distribution and classification confusion matrix

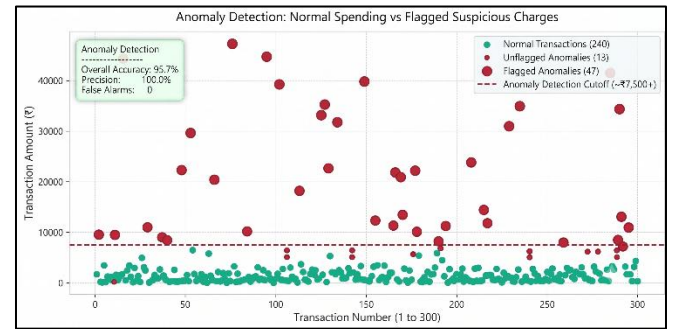


Fig. 3. Transaction spending distribution with flagged anomalies

C. Progressive Expense Forecasting

The Random Forest expenditure predictor was tested using an 83-day rolling walk-forward back-testing method, yielding 82.94% accuracy in prediction, a mean absolute error of ₹210.39 per day, a root mean squared error of ₹295.88, and an R^2 value of 0.5624. The directional accuracy (whether or not there will be a decrease or increase in expenditure) was relatively poor, standing at 48.78%, indicating that the algorithm is better at predicting the magnitude of spending than its direction. The average response latency was 44.86 ms.



Fig. 4. Actual versus AI-predicted daily expenses over a 30-day window

Overall, the results across all three modules – 100% rule alignment, 95.7% anomaly-detection accuracy and 82.9% forecast accuracy, are the evaluations of the performance of

distinct functional modules and, therefore, they cannot be aggregated into one common score. The scoring engine has faithfully used the tier rules and debt prevention methods provided in the design. The anomaly detection module has generated no false alarms; however, a recall rate of 78.33% proves that some anomaly transactions have been missed by the algorithm, which implies the need for the retraining and tuning of the threshold. The expense forecasting module has managed to meet all criteria related to its accuracy, mean error and R^2 , while the low rate of directional accuracy (48.78%) proves that the estimation of the volume of expenditures and forecasting the daily change of the spending are different abilities of the module, thus forecasts need to be interpreted together with their error criteria. Overall, the results prove that the explainability based on the rules and adaptive machine-learning can coexist in one personal finance management system. The scoring result proves that the predefined rules have been implemented correctly; however, the sensitivity analysis of the weights needs to be conducted.

V. CONCLUSION

This paper presented the Phase I financial-assessment layer of FinGear AI, comprising an adaptive, explainable Financial Health Score, a per-user category-specific anomaly-detection engine, and a progressive baseline-to-ARIMA-to-Random-Forest forecasting pipeline. Income-tier classification ensures that users with different financial circumstances are evaluated against suitable targets rather than a single generic rule, while explicit component-level formulae keep every score, alert and forecast traceable to interpretable evidence. Validation results — 100% alignment of the implementation with the predefined scoring rules, 95.67% anomaly-detection accuracy with 100% precision, and 82.94% forecasting accuracy — support the feasibility of this approach. This foundation is designed to support the later Digital Twin features of FinGear AI, including goal and budget planning, what-if simulation, AI-Copilot guidance and investment decision support.

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